Elham Esmaeilnia

Senior AI Engineer | LLM Applications, Intelligent Agents & AI Systems

Tehran, Iran | elham.e.shirvani@gmail.com | +98 911 128 7040
github.com/Elham-Esmaeilnia | linkedin.com/in/elham-esmaeilnia

PROFESSIONAL SUMMARY

Senior AI Engineer and researcher with experience spanning LLM applications, intelligent agents, Persian NLP, speech processing, deep learning, and data-intensive systems. Combines industry R&D experience with independent development of AI services, including LLM-based support-ticket triage and asynchronous voice-agent infrastructure. Experienced in designing reliable pipelines, semantic retrieval, structured outputs, and resilient backend services.

EXPERIENCES

Freelance AI Engineer | Independent Projects

Feb 2026 – Present

- Developing an LLM-driven support-triaging engine for automated ticket classification, semantic retrieval, and resilient processing.
- Building an asynchronous AI voice-sales-agent framework designed for low-latency interactions and reliable deployment.

Senior AI Engineer | Parnian Gostar

Aug 2025 – Jan 2026

- Designed and developed AlphaFusionNet, an AI-based financial modeling system.
- Implemented LLM-based intelligent agents for financial workflows.
- Developed time-series pipelines and deep-learning models.
- Contributed to AI-driven hybrid decision-making systems.

AI Engineer | Bertina

Dec 2024 – 2025

- Researched and developed Persian NLP systems, including text normalization and tokenization for grapheme-to-phoneme (G2P) workflows.
- Contributed as co-author to a G2P article from the Bertix Artificial Intelligence Group.
- Worked on language-understanding systems and a Persian diacritization model.

AI Research & Development Engineer | Farajahan

Nov 2023 – Dec 2024

- Developed intelligent question-answering and text-correction systems.
- Built a RAG-based system for generating responses grounded in Persian-language content.
- Developed a Persian speech-enhancement model and trained transformer-based language models.
- Created a multi-task Persian intelligent assistant using LLMs and CrewAI.

Big Data & Web Developer | Kanda Idea

Jun 2020 – Aug 2022

- Processed large-scale data using Spark, Hadoop, and HDFS; implemented analytics pipelines with Cassandra.
- Developed backend services, APIs, and web-based systems for IMIDRO and WEB MARINE.
- Designed use-case and sequence diagrams to support software architecture.

AI Researcher | IPM

Apr 2018 – Aug 2018

- Conducted research on deep learning and semi-supervised processing.

Instructor | Alavi Institute

Jun 2013 – Jul 2014

- Taught programming, computer fundamentals, and mathematics.

SELECTED PROJECTS

AI-Based Financial Portfolio Optimization System — https://github.com/Novoxpert/AlphaFusionNet
AI-based financial modeling and portfolio optimization work developed in the context of industry experience.

Persian NLP Pipeline: G2P, Tokenization & Text Correction
Persian-language NLP work covering grapheme-to-phoneme processing, text normalization, tokenization, and text correction.

Persian Diacritization Model
A model designed to add diacritics to unmarked Persian text.

RAG-Based Intelligent Assistant
An intelligent question-answering system that generates responses grounded in Persian-language content using retrieval-augmented generation.

Persian Speech Enhancement
Speech enhancement research and development using deep-learning methods.

Persian Multi-Task Intelligent Assistant
A Persian-language assistant leveraging LLMs and CrewAI to perform multiple task-oriented functions.

Web-Based Systems: IMIDRO & WEB MARINE
Web application and backend development delivered during the Big Data & Web Developer role.

Voice Sales Agent — https://github.com/Elham-Esmaeilnia/voice-sales-agent
A scalable AI voice-agent framework designed for low-latency sales interactions, using asynchronous patterns and a modular architecture for reliable deployment.

TECHNICAL SKILLS

Programming & Backend: Python, C++, MATLAB, JavaScript, HTML/CSS, FastAPI, REST APIs, WebSockets

AI / ML / Speech: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, OpenCV, Librosa, speech processing, deep learning, time-series modeling

LLMs & Agentic Systems: LLM applications, prompt engineering, RAG, LangChain, LangGraph, CrewAI, semantic retrieval, vector databases, structured outputs, tool calling

Data & Infrastructure: Spark, Hadoop, HDFS, Cassandra, MongoDB, Neo4j, Redis, Qdrant, FAISS, Docker, Docker Compose, Celery, RabbitMQ, Git

Automation & AI Development: n8n, MCP servers, Claude Code, Browser Use, LM Studio

Generative Image / Video Tools: Adobe Firefly, Stable Diffusion, Flux, Runway, Kling, Luma AI, Ideogram, Leonardo AI, Krea.ai, Freepik AI, Vidu, Hailuo AI, TensorArt, Fooocus, Renderforest, Dreamina, Grok,, InVideo AI, Shakker, PromptHero, Glif, Pixelcut, Upscayl, Vectorize.ai, PicWish, Stylar.ai, RenderNet, Apob, CGDream, Phygital, PowerPaint, LensGo, Decohere AI, AI Comic Factory, Dashtoon, Pika, Focal, EverArt, ZeoDepth, HyperBooth AI, DGBLOL, RTFlow.ai

EDUCATION

- PhD Candidate, Computer Engineering (AI & Robotics) — Islamic Azad University, Tehran Central
- MSc, Computer Engineering (Artificial Intelligence) — Rouzbehan University, Sari
- BSc, Computer Engineering — Shahid Beheshti University, Tehran

CERTIFICATIONS, HONORS & LANGUAGES

Certifications: Data Science & Python – DigiNext; Web Design – FanAsa; Teaching Skills – Ministry of Education; IT Internship – Agricultural Jihad Organization

Honors: National Mathematics Olympiad – Selected Participant; City-Level Mathematics Olympiad – Selected Participant; City-Level Physics Olympiad – Selected Participant

Languages: Persian (Native); English (Professional)